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AAS in AI Software Engineering

PY102

Welcome to PY102!

Today
Ron, welcome into PY102 🎯
This is where you start moving from “I can write code” to “I can design smart solutions.”

In this course, you’ll train your brain to think about:

how data is stored
how steps in a solution are organized
and how to tell if one solution is better than another as problems get bigger.
You don’t need to know any of that deeply yet—that’s what this course is for. You just need your Python basics and a bit of curiosity.

Let me give you the big picture of how the course is laid out.

Unit 1 – Foundations of Data Structures
You’ll look at how computers store and organize information in simple ways.
Main ideas:

Using Python lists as arrays
Accessing, updating, and removing data
Working with multi-dimensional “grids” of data (like a table or board)
Unit 2 – Stacks and Queues
Here you’ll see two very common ways to handle ordered data.
Main ideas:

What “Last-In-First-Out” (LIFO) means (stack behavior)
What “First-In-First-Out” (FIFO) means (queue behavior)
How these ideas show up in real life: undo history, browser history, lines at a store, etc.
Unit 3 – Algorithm Analysis
This unit is about why some solutions are faster or more efficient than others.
Main ideas:

Thinking in terms of “number of steps” an algorithm takes
How performance changes as data grows
A gentle intro to Big O notation as a simple language for talking about efficiency
Unit 4 – Searching and Sorting
These are core tools you’ll use everywhere as a programmer.
Main ideas:

How to search through data (linear search vs. binary search)
How to sort data (bubble sort, insertion sort, and Python’s built-in tools)
Comparing these approaches at a beginner-friendly level
Now, how you’ll be graded.

Each unit ends with a review of that unit’s material.
The last unit ends with a big final review that covers the full course.

Grading breakdown:

Reviews 1–3:
worth 10% each → 30% total
not timed
unlimited retakes
Last unit review:
worth 60% of your grade
timed
only one retake

Weekly discussions:
10% of your grade
based on participating in weekly discussions until the term end date
You can also earn up to 10 bonus points if you submit and pass all reviews on time.

Review format (what they feel like):

Mix of multiple-choice and open-ended questions
Focus on:
explaining ideas in your own words
reading code and saying what it does
predicting outputs or errors
reasoning about how efficient a small algorithm is
debugging and recognizing common patterns
Important policy note:
For this course, you are not allowed to use external AI tools on any review questions.
The reviews are meant to measure your understanding and thinking.

If you ever need official or super detailed info about grading, policies, or timing, reach out to Student Services—they're the authority on that.

From my side: you've now got what you need for this orientation.
You understand the structure, units, and how reviews and participation fit into your grade.

Want to jump into the next lesson now, or do you want to pause and ask anything about how the course is set up?

How would you rate this lesson?